Component Optimization

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#react

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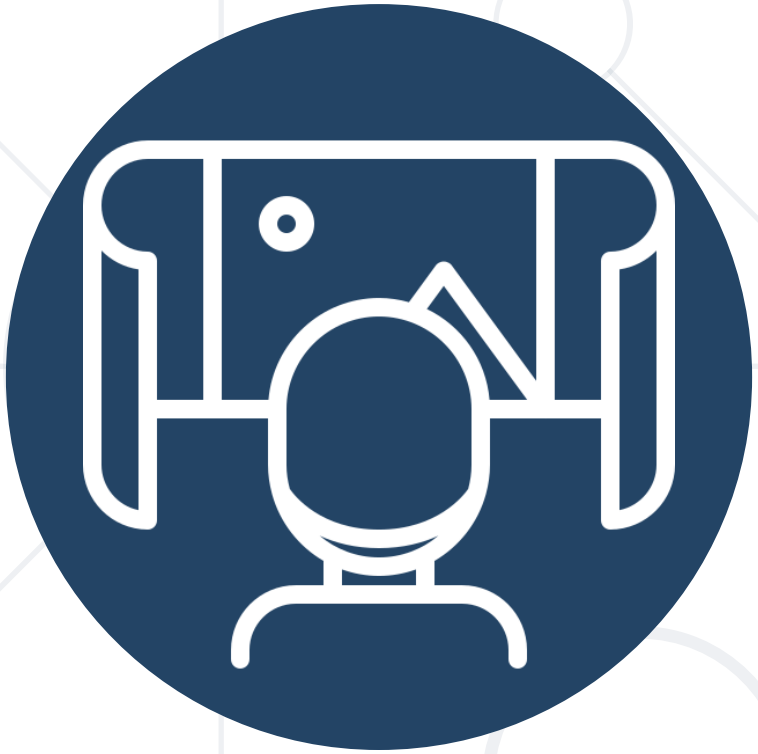
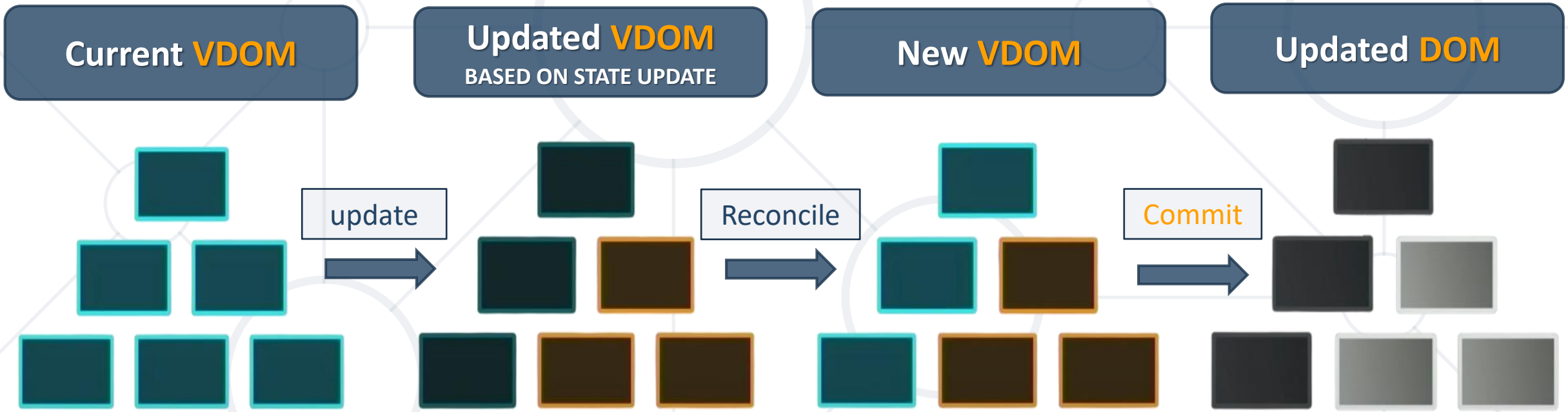


Virtual DOM

Flow


Current VDOM
Updated VDOM
BASED ON STATE UPDATE
New VDOM
Updated DOM
update
Reconcile
Commit

Profiler

Profiler

- <**Profiler**> lets you measure rendering performance of a React tree programmatically
- Helps identify **performance** bottlenecks and optimize rendering efficiency
  - Wrap a **component tree** with <Profiler> to measure its rendering performance

```
<Profiler id="App" onRender={onRender}>
  <App />
</Profiler>
```

Profiler Props

- A string identifying the part of the UI you are measuring (**id**)
- Callback function executed when components in the profiled tree update (**onRender**)
- Wrap a **component tree** with <Profiler> to measure its rendering performance

```
<Profiler id="App" onRender={onRender}>
  <App />
</Profiler>
```

onRender

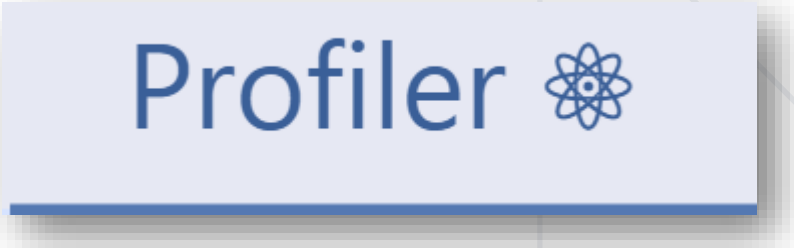
- React calls **onRender** with information about each render cycle
  - **id**: The id prop of <Profiler>
  - **phase**: "mount", "update", or "nested-update"
  - **actualDuration**: Time (ms) spent rendering the profiled tree
  - **baseDuration**: Estimated time to render the entire subtree without optimizations
  - **startTime**: Timestamp when rendering began
  - **commitTime**: Timestamp when React committed the update

React Developer Tools Profiler

- A built-in feature of **React DevTools** for interactive performance profiling
  - Shows component render **durations** and commit times
  - Highlights components that rendered during each **commit**
  - Displays flame **graphs** and ranked **views** for performance insights

React Developer Tools Profiler Use

- Open React Developer Tools in your **browser**
- Navigate to the **Profiler** tab
- Click the **Record** button and interact with your application
- Stop recording to analyze render performance
- Use the flame graph to identify costly re-renders



React.memo

memo

- **memo** lets you skip re-rendering a component when its props are unchanged
- Wrap a component in **memo** to get a **memoized** version of that component
- **returns** a new React component
- It behaves the same as the component provided to memo except that React will not always re-render it when its parent is being re-rendered unless its props have **changed**

memo

- **Memoized** components skip re-renders during parent updates if their new props are the same as the old props

```
const Child = React.memo(function Child({ data }) {
      console.log("Child re-renders");
      // Only logs if `data` changes
return
      <div>{data}</div>;
 });
```

useCallback

useCallback

- **useCallback** lets you cache a a function between re-renders, returning the same function instance unless dependencies change
- **Prevents** unnecessary re-renders in child components (e.g., when passing callbacks to React.memo components)
- Dependency array must **include** all variables used inside the function

useCallback

```
const Parent = () => {
 const [count, setCount] = useState(0);
 // Without useCallback, `handleClick` is recreated on every render:
const handleClick = useCallback(() => {
 setCount(c => c + 1); }
, []); // Same function unless dependencies change
return <Child onClick={handleClick} />;
 };
 const Child = React.memo(({ onClick }) => (
 <button onClick={onClick}>Click</button>
));
```

useMemo

useMemo

- **useMemo** lets you cache the result of a calculation between re-renders, re-calculating it only when dependencies change
- Avoids **expensive** recalculations on every render
- Prevents unnecessary re-renders in child components when passing non-primitive values (e.g., objects, arrays)
- Used for **costly** computations (e.g., filtering/sorting large lists)

useMemo

```
const List = ({ items, filterTerm }) => {
const filteredList = useMemo(() => {
 return
items.filter(item => item.includes(filterTerm));
//Only re-runs if `items` or `filterTerm` changes
}, [items, filterTerm]);

return <Child list={filteredList} />;
 };

const Child = React.memo(({ list }) => ( /* ... */ ));
```

Compiler

React Compiler

- A **compiler** tool that automatically optimizes React apps by
  - **Auto-memoizing** components, values, and functions
  - Eliminating unnecessary re-renders (like manual **useMemo**/**useCallback**)
  - Reducing bundle size via dead code removal
- Still **experimental**

How to Use the React Compiler

- Install packages

```
npm install babel-plugin-react-compiler
```

- Add the compiler plugin to your Babel config (e.g., babel.config.js)

```
module.exports = {
      plugins: [ ['babel-plugin-react-compiler’],
// ...other plugins
      ],
 };
```

- Enable Strict Mode

How to Use the React Compiler

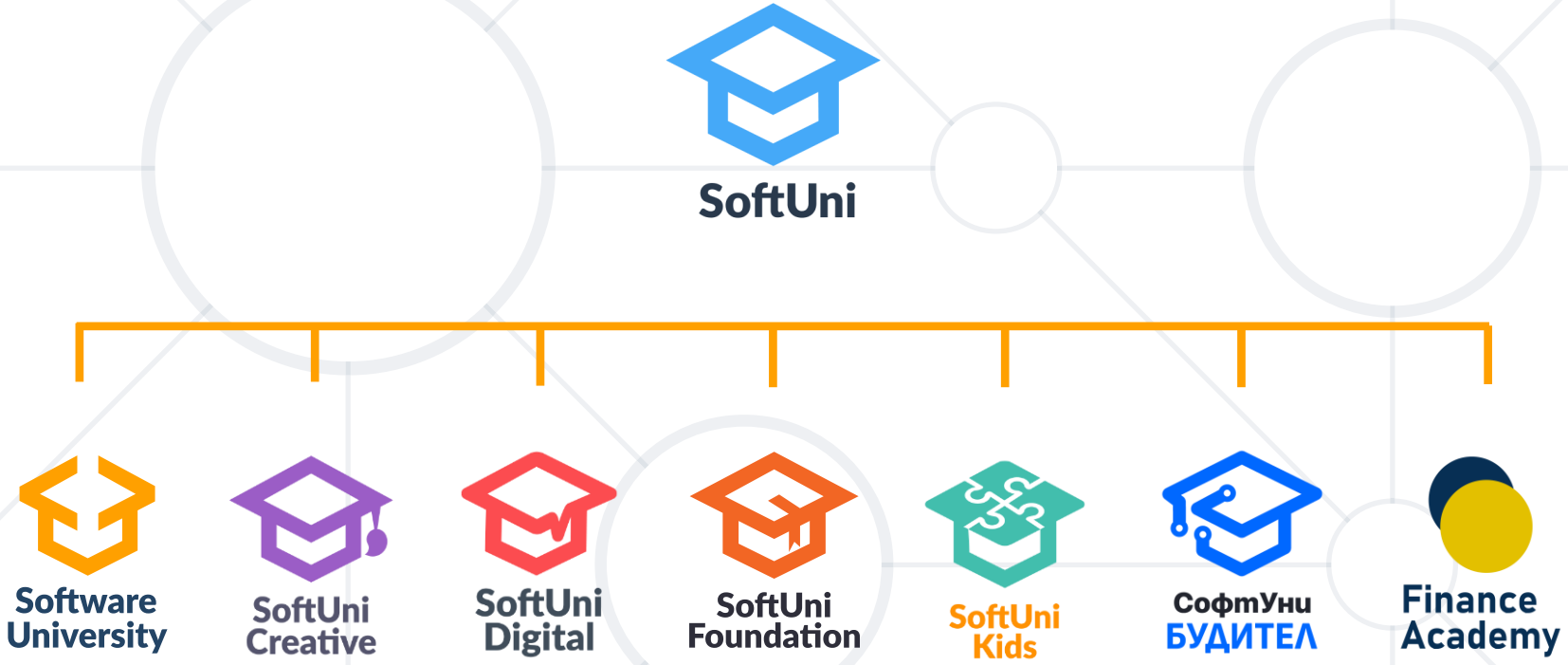
- The compiler automatically optimizes components, values, and functions

```
// No need for manual `useMemo`/`useCallback`!
 const MyComponent = ({ items }) => {
 const filteredList = items.filter(item => item.active);
//Auto-memoized
return
 <Child list={filteredList} />;
};
```

Summary

- **Profiler** find performance bottlenecks
- **React.memo** prevent re-renders of functional components if props are unchanged
- **useCallback** lets you cache a function definition between re-renders
- **useMemo** lets you cache the result of a calculation between re-renders
- **Compiler** replaces manual useMemo/useCallback, but is experimental for now

Questions?



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